

CS305 Database Systems

Lecture 18: SQL

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Outline
Relational model
Keys
SQL queries
Joins
Normalization
Transactions

The Relational Model

A relation is a table of rows and columns.

Each row is a tuple representing one entity.

Each column is an attribute with a domain.

Primary Key

A primary key uniquely identifies each row in a table.

It cannot contain NULL values.

Every table should declare one.

Foreign Key

A foreign key references the primary key of another table.

It enforces referential integrity between relations.

SELECT Statement

SELECT retrieves rows from one or more tables.

The WHERE clause filters rows by a predicate.

Projection

Projection chooses which columns appear in the result.

It is written as the column list after `SELECT`.

Aggregate Functions

Aggregate functions compute a single value over many rows.

COUNT returns the number of rows in a group.

GROUP BY

GROUP BY partitions rows into groups sharing a value.

Aggregates are then computed per group.

HAVING

HAVING filters groups after aggregation.

WHERE filters rows before grouping.

Inner Join

An inner join returns rows matching in both tables.

The join condition compares a foreign key to a primary key.

Outer Join

A left outer join keeps unmatched rows from the left table.

Missing values are padded with NULL.

Subqueries

A subquery is a query nested inside another query.

A correlated subquery references the outer query.

Views

A view is a named query stored in the schema.
It presents derived data as if it were a table.

Figure 3: ER diagram

Indexes

An index is a data structure that speeds up lookups.
It trades extra write cost for faster reads.

Functional Dependency

A functional dependency means one attribute determines another.

It is written $X \rightarrow Y$.

First Normal Form

First normal form requires every attribute to be atomic.

Repeating groups are not allowed.

Second Normal Form

Second normal form removes partial dependencies on a composite key.

Every non-key attribute depends on the whole key.

Third Normal Form

Third normal form removes transitive dependencies.

Non-key attributes must not determine other non-key attributes.

Denormalization

Denormalization reintroduces redundancy to improve read speed. It is a deliberate trade-off against update anomalies.

Transactions

A transaction is a unit of work executed atomically.
It either commits entirely or aborts entirely.

ACID Properties

Atomicity means all operations succeed or none do.

Durability means committed changes survive failure.

Isolation Levels

Isolation determines how concurrent transactions see each other.

Serializable is the strictest level.

Deadlock

A deadlock occurs when transactions wait on each other forever.

The system resolves it by aborting one transaction.

Stored Procedures

A stored procedure is program logic stored in the database. It reduces round trips between client and server.

Triggers

A trigger executes automatically in response to a table event.

It can enforce rules that constraints cannot express.

A trigger runs due to changes in the table data.

Query Optimization

The optimizer chooses a low cost execution plan.

Cost is estimated from table statistics.

Figure 7: query plan tree

Summary

Keys, joins, normalization, transactions

References

Silberschatz, Database System Concepts

Appendix
Additional reading